

PROJECT REPORT

Quality Prediction in a Mining Process

Submitted By

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Company

Universal Consulting Technologies (UCT)

1. Introduction

Universal Consulting Technologies (UCT) is a technology-driven organization specializing in Artificial Intelligence, Machine Learning, Data Analytics, and Industry 4.0 solutions. As part of the internship program, a real-world industrial manufacturing project titled "**Quality Prediction in a Mining Process**" was undertaken.

The project focuses on predicting the percentage of silica concentrate in an iron ore flotation plant using machine learning techniques. The ability to predict silica levels in advance enables mining engineers to make proactive decisions that improve production quality and reduce operational losses.

2. Background of the Project

Mining industries use flotation plants to separate valuable minerals from unwanted materials. The quality of the final concentrate is measured by the percentage of silica impurity present in the ore.

Traditionally, laboratory measurements of silica concentration are available only after a delay. This delay can result in inefficient process adjustments and increased waste generation.

Using machine learning, it is possible to analyze historical process data and predict silica concentration in advance, allowing engineers to optimize the flotation process before quality degradation occurs.

3. Problem Statement

The primary objective of this project is to develop a machine learning model capable of predicting the **% Silica Concentrate** in a flotation plant using operational process parameters.

Key Questions Addressed

- Is it possible to predict % Silica Concentrate accurately?

- Can silica concentration be predicted ahead of time?
 - Can predictive analytics help engineers reduce waste and improve efficiency?
 - Can the model provide early warnings when silica levels exceed acceptable limits?
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4. Relevance of the Project

This project has significant industrial relevance because:

- Improves ore quality prediction.
 - Reduces tailings and production waste.
 - Supports proactive decision-making.
 - Enhances operational efficiency.
 - Contributes to environmental sustainability.
 - Demonstrates practical application of Artificial Intelligence in manufacturing.
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5. Dataset Description

The dataset contains operational data collected from an iron ore flotation plant.

Major Features

- Date and Time
- % Iron Feed
- % Silica Feed
- Starch Flow
- Amina Flow
- Ore Pulp Flow
- Ore Pulp pH
- Ore Pulp Density
- Flotation Column Air Flow Measurements
- Flotation Column Level Measurements
- % Iron Concentrate
- % Silica Concentrate (Target Variable)

The target variable selected for prediction is:

% Silica Concentrate

6. System Design

Workflow

Data Collection ↓ Data Cleaning ↓ Data Preprocessing ↓ Feature Selection ↓ Model Training (XGBoost) ↓ Prediction ↓ Performance Evaluation ↓ Early Warning System

Selected Model

XGBoost Regressor

The XGBoost algorithm was selected because it provides high prediction accuracy, handles large datasets efficiently, and performs well with industrial process data.

7. Implementation Details

Software Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn
- XGBoost

Data Cleaning

The following preprocessing activities were performed:

- Removal of unwanted spaces from column names.
- Conversion of date column into datetime format.
- Handling missing values using forward fill.
- Conversion of process variables into numerical format.
- Sorting records based on timestamp.

Model Development

The dataset was divided into:

- Training Data (80%)
- Testing Data (20%)

The XGBoost Regressor model was trained using process variables to predict silica concentration.

8. Results

The model successfully predicted silica concentration using historical flotation plant data.

Outputs Generated

- Predicted % Silica Concentrate
- Actual vs Predicted Comparison
- Performance Metrics

- Future Silica Prediction
- Early Warning Alerts

Performance Metrics

The following metrics were used:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- R² Score

Visualization

Graphs generated include:

- Actual vs Predicted Silica Concentration
- Prediction Error Distribution
- Feature Importance Analysis

These visualizations helped evaluate model performance and understand the contribution of process variables.

9. GitHub Repository

The complete project source code, dataset handling scripts, machine learning model implementation, and visualizations were maintained using GitHub for version control and project management.

GitHub was used to:

- Store project code.
 - Track modifications.
 - Maintain documentation.
 - Facilitate collaboration and reproducibility.
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10. Expected Outcome

The developed system is capable of:

- Predicting silica impurity in advance.
- Assisting engineers in operational decision-making.
- Reducing production losses and tailings.
- Improving mining efficiency.
- Supporting environmental sustainability initiatives.

The system can function as an intelligent monitoring and early warning solution for flotation plant operations.

11. Learnings from the Project

Through this project, the following skills and concepts were learned:

Technical Learnings

- Data Cleaning and Preprocessing
- Exploratory Data Analysis
- Machine Learning for Regression
- XGBoost Algorithm
- Industrial Data Analytics
- Data Visualization
- Performance Evaluation Metrics

Professional Learnings

- Real-world industrial problem solving
- Data-driven decision making
- Project documentation
- GitHub version control
- Model development lifecycle

12. Conclusion

The Quality Prediction in a Mining Process project successfully demonstrated the application of Machine Learning in industrial manufacturing. Using historical flotation plant data, the developed XGBoost model predicted silica concentration with high accuracy.

The project highlights how predictive analytics can improve ore quality, reduce waste, enhance operational efficiency, and support sustainable mining practices. This internship project provided valuable exposure to real-world industrial AI applications and practical machine learning implementation.